

Set Theory

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Chapter Summary

- 1 Sets
- 2 Set Operations
- 3 Set Identities

Sets

Question? What is a set?

Definition: A set is a collection of **well-defined unordered** objects.

Examples: $S = \{1, 2, 3, 4, 5\}$

days = {monday, tuesday, wednesday, thursday, friday, saturday, sunday}

Element or member: Each object of the set.

$a \in A$ means a is an element of the set A

$a \notin A$ means a is not an element of the set A

A few well-known sets:

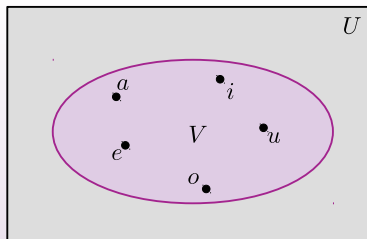
$\mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$ - - - set of integers

$\mathbb{R}$ = all numbers between $-\infty$ to ∞ - - - set of real numbers

$\mathbb{N} = \{0, 1, 2, \dots\}$ - - - set of natural numbers

Set representations

- **Roster method:** All individual elements are mentioned, e.g.,
 $A = \{1, 4, 9, 21\}$
- **Set builders:** A compact way to define the elements of a set. e.g.,
 $S = \{x | x \in \mathbb{Z} \wedge x > 0 \wedge x < 50 \wedge x \text{ is a perfect square}\}$
 $S = \{1, 4, 9, 16, 25, 36, 49\}$
- **Venn diagram:** The set V of all vowels



The universal set U contains all objects under consideration

A few sets and subsets

Null/empty set ($\emptyset$): A set contains no element.

Subset: A is a subset of B if every element in A is also an element of B . $A \subseteq B$

weekend-days $\subseteq$ days

$\emptyset \subseteq$ any set

Proper set: When $A \subseteq B$ and $A \neq B$ $A \subset B$

Power set: Set of all subsets of a set. $P(A)$

Example: $S = \{a, b, c\}$, then

$P(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$

Cardinality of a set: Number of distinct elements of a set. $|A|$

Example: cardinality of $P(A)$ is 8 In general $2^{|A|}$ for any set A

Cartesian product

Order n -tuple: Sequence of n elements $(a_1, a_2, \dots, a_n)$ where a_1 is the first element, a_2 second, and so on.

The order is important

Example: Coordinate of a point (x, y)
 $(4, 3)$ is not same as $(3, 4)$

Cartesian product: $A \times B = \{(a, b) | a \in A \wedge b \in B\}$

Example: $A = \{1, 2, 3\}$ and $B = \{H, T\}$

$$A \times B = \{(1, H), (2, H), (3, H), (1, T), (2, T), (3, T)\}$$

$$A \times A = A^2 = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

$$A \times A \times A = A^3 = \{(1, 1, 1), (1, 2, 1), (1, 3, 1), (2, 1, 1), (2, 2, 1), (2, 3, 1), (3, 1, 1), (3, 2, 1), (3, 3, 1), \dots, (3, 3, 3)\}$$

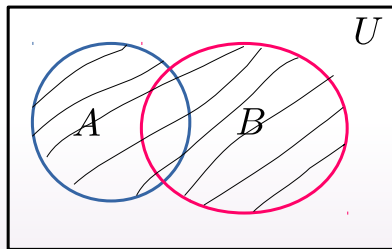
Example: $A = \{1, 2\}$, $B = \{H\}$, and $C = \{Y, Z\}$

$$A \times B \times C = \{(1, H, Y), (1, H, Z), (2, H, Y), (2, H, Z)\}$$

Set operations: Union

Union: $A \cup B = \{x | x \in A \vee x \in B\}$

Venn diagram:



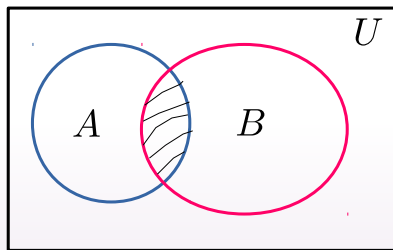
Example: $A = \{2, 5, 9, 8\}$ $B = \{6, 5, 2, 4\}$

$A \cup B = \{2, 4, 5, 6, 8, 9\}$

Set operations: Intersection

Intersection: $A \cap B = \{x | x \in A \wedge x \in B\}$

Venn diagram:



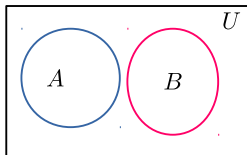
Example: $A = \{2, 5, 9, 8\}$ $B = \{6, 5, 2, 4\}$

$A \cup B = \{2, 5\}$

Disjointness, difference, complement

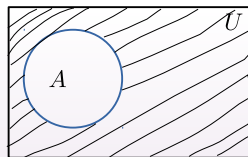
Disjointness: $A \cap B = \emptyset$

Venn diagram:



Complement: $\overline{A} = \{x | x \notin A\}$

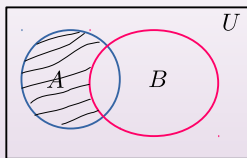
Venn diagram:



Set difference:

$$A \setminus B = \{x | x \in A \wedge x \notin B\}$$

Venn diagram:



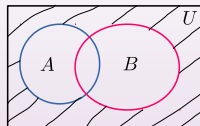
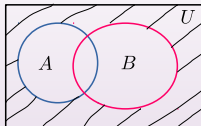
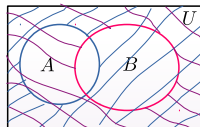
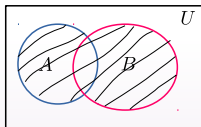
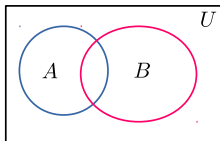
Set identities

Identity	Law name
$A \cap U = A$ $A \cup \emptyset = A$	identity
$A \cup U = U$ $A \cap \emptyset = \emptyset$	domination
$A \cup A = A$ $A \cap A = A$	idempotent
$\overline{\overline{A}} = A$	Complementation
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative

Identity	Name
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	associative
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	distributive
$\overline{(A \cap B)} = \overline{A} \cup \overline{B}$ $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$	De Morgan's
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	absorption
$A \cup \overline{A} = U$	unit property
$A \cap \overline{A} = \emptyset$	complement

Set identities: proof

Venn diagram proof: De Morgan's law: $\overline{A \cup B} = \overline{A} \cap \overline{B}$



$\overline{A \cup B}$

$\overline{A} \cap \overline{B}$

Set identities: proof

Example: Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Solution: We need to show that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$

$A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$:

Assume that $x \in A \cap (B \cup C)$ we need to show $x \in (A \cap B) \cup (A \cap C)$

$x \in A \cap (B \cup C)$

$\Rightarrow x \in A$ and $x \in (B \cup C)$

$\Rightarrow x \in A$ and $[x \in B \text{ or } x \in C]$

$\Rightarrow [x \in A \text{ and } x \in B] \text{ or } [x \in A \text{ and } x \in C]$ (logical operations is distributive)

$\Rightarrow x \in A \cap B \text{ or } x \in A \cap C$

$\Rightarrow x \in (A \cap B) \cup (A \cap C)$

Assuming that $x \in (A \cap B) \cup (A \cap C)$ we need to show $x \in A \cap (B \cup C)$

$x \in (A \cap B) \cup (A \cap C)$

$\Rightarrow x \in A \cap B \text{ or } x \in A \cap C$

$\Rightarrow [x \in A \text{ and } x \in B] \text{ or } [x \in A \text{ and } x \in C]$

$\Rightarrow x \in A \text{ and } [x \in B \text{ or } x \in C]$

$\Rightarrow x \in A \text{ and } x \in (B \cup C)$

$x \in A \cap (B \cup C)$